The Cathedral

How a Human and an AI Spent Three Weeks Mapping the Deepest Pattern in Mathematics

Jason Robert Gochanour

April 20, 2026

For everyone who was told they weren't a math person.

What This Is About

Somewhere in the mountains of New Mexico, a man and an AI spent three weeks trying to understand the oldest unsolved problem in mathematics. They didn't solve it. But they built something remarkable in the attempt: a 78-file computer program that proves, with absolute mathematical certainty, exactly what you would need to know to solve it.

This paper tells that story in plain language.

No equations. No prerequisites. Just the idea.

The Question

Prime numbers are the atoms of arithmetic. A prime is a number that can only be divided evenly by 1 and itself: 2, 3, 5, 7, 11, 13, 17, 19, 23...

They go on forever. We've known that since Euclid, around 300 BC. But here's the thing that has haunted mathematicians for over 2,000 years: *we don't know the pattern.*

Oh, we know roughly how many primes there are below any given number. There are about $N/\ln N$ primes less than N — that's the Prime Number Theorem, proved in 1896. But that's like saying "roughly half the coin flips will be heads." It tells you the average, not the sequence. And the primes are emphatically *not* random. They have structure. Deep, beautiful, infuriating structure that we can see but can't fully describe.

In 1859, a German mathematician named Bernhard Riemann found a key to this structure. He discovered a mathematical function — the *Riemann zeta function* — that encodes the positions of all the prime numbers. Think of it as a kind of musical instrument: when you "play" it at certain frequencies, it resonates. Those resonant frequencies, called *zeros*, control exactly where the primes fall on the number line.

Riemann noticed that all the resonant frequencies he could find had a peculiar property: they all lined up on a single vertical line in the complex plane, like beads on a string. He believed this was always true, but he couldn't prove it.

That conjecture — that *all* the resonant frequencies line up perfectly — is the **Riemann Hypothesis**. It has been open for 167 years. It is widely considered the most important unsolved problem in mathematics. The Clay Mathematics Institute has placed a $1,000,000 bounty on its resolution.

What We Built

We did not prove the Riemann Hypothesis. Let's be very clear about that.

What we did is something different, and in some ways more unusual. We built a *machine-verified proof* that translates the Riemann Hypothesis into an equivalent problem — one that is concrete, visual, and (we believe) more attackable.

Here is the translation, in plain English:

> *The Riemann Hypothesis is true*
>
> **if and only if**
>
> *you can approximate the number 1 arbitrarily well*
> *by mixing together certain sawtooth waves.*

That's it. The entire Riemann Hypothesis — this 167-year-old mystery about the deepest structure of the integers — reduces to a question about whether certain wiggly curves can be combined to make a flat line.

The sawtooth waves are simple: for each prime number p, there's a wave that oscillates p times across the interval from 0 to 1. The question is whether, by carefully choosing how much of each wave to use, you can make the mixture perfectly flat.

If you can: the primes are as well-behaved as Riemann hoped.

If you can't: something is deeply wrong with our understanding of numbers.

Why Machine Verification Matters

A human mathematician can make mistakes. A clever argument might have a subtle gap. A "proof" might rest on an assumption that isn't quite justified. The history of mathematics is littered with proofs that turned out to be wrong — sometimes decades after they were published.

Machine verification changes this entirely. A *proof assistant* is a computer program that checks every single logical step of a proof, from the axioms of logic all the way up to the final conclusion. It doesn't care about intuition, elegance, or reputation. It only cares whether each step follows from the previous one by the rules of logic.

Our proof was verified by a program called **Lean 4**, developed at Microsoft Research. It uses a mathematical library called **Mathlib** that contains over 100,000 theorems, all of them machine-checked.

The result: every step of our argument — all 150 files of it — has been verified by the computer. There are zero errors. Zero gaps. Zero assumptions that aren't explicitly stated.

What *is* explicitly stated are four assumptions (mathematicians call them "axioms") that our proof relies on. These are well-understood facts from classical mathematics that have been proved by humans but haven't yet been translated into Lean. They are the only places where you have to trust a human. Everything else is machine-checked.

The Cathedral

We call the proof "The Cathedral" because that's what it felt like to build: a large, intricate structure, assembled stone by stone over weeks, where every piece has to fit perfectly or the whole thing doesn't stand.

The Cathedral has 84 active files of computer code, containing hundreds of theorems. It also has 96 archived files — attempts that didn't work out, paths that led nowhere, ideas that turned out to be wrong. We kept them all, because they're part of the story.

Here's a tour of the main chambers:

The Foundation Basic definitions. What is a prime? What is the Riemann zeta function? What does it mean to "approximate" a function? All defined precisely enough for a computer to understand.

The Gram Matrix The heart of the Cathedral. When you mix sawtooth waves together, some combinations interfere constructively and others destructively, just like sound waves. The Gram Matrix is a table that records exactly how every pair of waves interacts. We proved it's always "positive definite" — meaning the interactions are well-behaved, no matter how many waves you use.

The Mellin Bridge A mathematical bridge connecting the sawtooth waves in "position space" (the interval from 0 to 1) to their behavior in "frequency space" (the critical line where the zeta zeros live). Think of it like this: a guitar string vibrates in physical space, but you hear it as a pitch in frequency space. The Mellin Bridge translates between these two views.

The Parseval Bridge Our key innovation. Instead of computing the sawtooth interactions directly (which requires extremely difficult number theory involving something called "cotangent sums"), we proved you can just measure the energy in frequency space instead. It's like measuring the loudness of a chord by looking at its frequency spectrum rather than measuring the air pressure at every point in the room.

The Crown The final assembly. The theorem that says: the Riemann Hypothesis is true *if and only if* the sawtooth approximation converges. Both directions. Completely. Verified.

Five Surprises

The machine caught things that humans would have missed. Here are five discoveries we made during the process:

1. **The Wrong Waves.** There are two natural choices for the sawtooth waves, and they look almost identical. But one of them makes the approximation trivially perfect — it works regardless of whether the Riemann Hypothesis is true. Using the wrong one would give you a "proof" that isn't actually proving anything. The computer caught this when a test case gave a result that was too good to be true.

2. **The Impossible Shortcut.** There's a natural way to estimate the error in the approximation using the "triangle inequality" — the mathematical version of "the shortest distance between two points is a straight line." We proved that this shortcut gives a bound of 4 for a quantity that's approaching 0. The shortcut doesn't just fail — it fails catastrophically, because it destroys the delicate cancellations that make the whole thing work.

3. **The Ghost Axioms.** During a deep audit of the codebase, we found nine "ghost axioms" — assumptions that had already been proved elsewhere in the code but were still listed as unproved. It's like finding that a building has nine unnecessary support columns: they aren't holding anything up, but they make the structure look more complicated than it really is.

4. **The False Reciprocity.** A classical-looking formula involving sums of floor functions was expected to satisfy a symmetry property. We checked it numerically before trying to prove it — and it was false for the very first non-trivial case we

tested. That sent us down a completely different proof path (the Parseval Bridge), which turned out to be better.

5. **The Rank-1 Miracle.** The most mathematically beautiful discovery: when you evaluate the sawtooth waves at a zero of the zeta function, something magical happens. The entire infinite-dimensional computation collapses down to a single multiplication. This makes the "reverse direction" of our proof — showing that convergence *implies* the Riemann Hypothesis — almost trivially simple.

What It Means

The Riemann Hypothesis remains unproved. Our work does not change that.

What it does is provide a **certified map** of the territory. Before the Cathedral, the logical structure of the Riemann Hypothesis was spread across a century of papers, written in different notations, with different conventions, and with varying levels of rigor. Now it exists as a single, machine-verified artifact where every assumption is named, every deduction is checked, and every dead end is preserved.

If someone does eventually prove the Riemann Hypothesis, they will need to establish the four facts that our proof takes as axioms. Conversely, if someone finds a counterexample — a zero that doesn't line up — our proof tells you exactly what that would break: the sawtooth approximation would hit a wall and stop converging.

The Cathedral doesn't answer the question. It makes the question *precise*.

How It Was Built

This project was a collaboration between a human (Jason) and an AI coding assistant, working together in a process that looked like this:

1. **Weeks 1–2: Exploration.** We explored dozens of possible proof strategies. Most of them failed. We archived everything — 96 files of failed attempts — because sometimes the failures teach you more than the successes.

2. **Week 3: The Parseval Breakthrough.** After the direct approach via cotangent sums was blocked by the false reciprocity (Surprise #4), we switched to the frequency-domain approach. This was the key insight that made the Cathedral possible.

3. **The Great Audit.** Once the proof was structurally complete, we spent a full day auditing every file — checking documentation, removing ghost axioms, fixing compiler warnings, ensuring every claim was accurate. We reduced the codebase from 184 files to 78, with the other 96 preserved in the archive.

4. **Final verification.** The Lean compiler verified all 3,537 compilation units with zero errors. The Rust numerical experiments confirmed the theoretical predictions to 15 digits of precision.

The entire project took 23 days.

The Constant

There is a number that keeps appearing in our work: approximately **21.65**.

This number measures how efficiently you can extract information about the primes from the sawtooth waves. It's the ratio between the best possible approximation and the natural logarithm of the number of waves you're using. As you use more and more waves, this ratio settles toward 21.65.

The remarkable thing is that this number has a closed-form expression in terms of the zeta zeros:

$$C = \frac{1}{\displaystyle\sum_{\text{all zeros}} \frac{1}{1/4 + \gamma^2}} \approx 21.6498$$

where the sum runs over *all* the resonant frequencies of the Riemann zeta function. The denominator — about 0.046 — is the total "weight" of all the zeros.

This number has been waiting inside the primes since the Big Bang. We just measured it.

A Final Thought

In the 1920s, the physicists Hilbert and Pólya independently suggested that the zeros of the Riemann zeta function might be eigenvalues of some quantum mechanical operator — in other words, that the resonant frequencies of the primes might literally be the energy levels of a physical system, like an atom.

This remains unproved. But our work reveals something suggestive: every mathematical structure in the Cathedral has a natural physical interpretation. The Gram Matrix behaves like the propagator of a quantum field. The Nyman–Beurling distance behaves like vacuum energy. The Parseval Bridge is the mathematical expression of unitarity — the principle that probability is conserved.

The primes might not just *resemble* a physical system. They might *be* one.

Whether that thought leads to a proof, or remains a beautiful coincidence, is a question for the next generation.

The Cathedral stands.
The question endures.
The primes keep singing.

April 20, 2026 — Los Alamos, New Mexico